

6-3.

$$F_3 \cdot a - F_1 \cdot a = F_e \cdot$$

$$F_2 a - F_4 a = 0$$

$$\text{设中心点 } O \quad \Delta l_2 = \Delta l_4 = \Delta l_0.$$

因为板是刚性，故会产生一个固定转动角 θ .

$$\Delta l_1 = \Delta l_0 - \frac{\sqrt{2}}{2} a \cdot \theta$$

$$\Delta l_3 = \Delta l_0 + \frac{\sqrt{2}}{2} a \cdot \theta.$$

$$\text{由胡克定律 } F = k \Delta x \quad \therefore F_1 = k(\Delta l_0 - \frac{\sqrt{2}}{2} a \theta) \quad F_2 = k \Delta l_0 \quad F_3 = k(\Delta l_0 + \frac{\sqrt{2}}{2} a \theta) \\ F_4 = k \Delta l_0$$

$$\text{代入平衡方程 } k(4\Delta l_0) = F$$

$$\therefore \sqrt{2} a \theta = 4\Delta l_0 e.$$

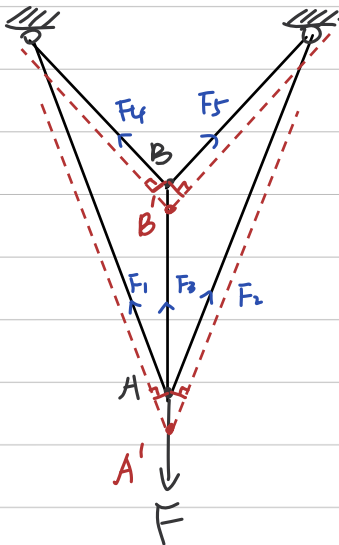
$$\text{求解得到 } F_1 = \frac{F}{4} - \frac{\sqrt{2} F_e}{2 a}$$

$$F_2 = \frac{F}{4}$$

$$F_3 = \frac{F}{4} + \frac{\sqrt{2} F_e}{2 a}$$

$$F_4 = \frac{F}{4}.$$

6-4.



$$\text{对A点: } F = F_3 + 2F_1 \cos 30^\circ = F_3 + \sqrt{3} F_1$$

$$\Delta l_1 = \Delta l_2 \quad \Delta l_A \cdot \frac{\sqrt{3}}{2} = \Delta l_1 \quad \frac{1}{2} \Delta l_B = \Delta l_4 = \Delta l_5.$$

$$= \frac{F_1 \sqrt{3} l}{EA} \quad \Delta l_A - \Delta l_B = \frac{F_3 l}{EA} = \frac{F_4 l}{EA}.$$

$$\text{对B点: } F_3 = F_4$$

$$-2\Delta l_4 + \frac{2}{\sqrt{3}} \Delta l_1 = \frac{F_3 l}{EA}$$

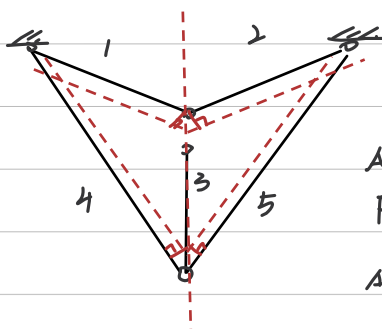
$$-\frac{2F_4 l}{EA} + \frac{2}{\sqrt{3}} \cdot \frac{F_1 \sqrt{3} l}{EA} = \frac{F_3 l}{EA}.$$

$$\therefore -2F_4 + 2F_1 = F_3$$

$$\therefore F_1 = \frac{3}{2 + \sqrt{3}} F \quad F_3 = \frac{2\sqrt{3}}{2 + \sqrt{3}} F.$$

$$\therefore \Delta l_1 = \frac{\sqrt{3} F_1 l}{EA} = \frac{3\sqrt{3}}{2 + \sqrt{3}} \frac{F l}{EA}$$

$$\therefore \Delta l_A = \frac{2}{\sqrt{3}} \Delta l_1 = \frac{6}{2 + \sqrt{3}} \frac{F l}{EA}$$



由分析得. 1. 2. 受拉. 4. 5 受压.

$$\Delta l_3 = \Delta - \frac{\Delta l_1}{\cos 60^\circ} - \frac{\Delta l_4}{\cos 30^\circ}$$

$$F_4 = F_5 = \frac{\sqrt{3}}{2} F_3 \quad F_1 = F_2 = F_3$$

$$\Delta l_1 = \frac{F_1 l_1}{EA} \quad \Delta l_4 = \frac{F_4 l_4}{EA} \quad \Delta l_3 = \frac{F_3 l_3}{EA}$$

$$\therefore F_3 = 0.241 \frac{EA \Delta}{l} = F_1 = F_2$$

$$F_4 = F_5 = 0.139 \frac{EA \Delta}{l}$$